

Rishi Gupta *Senior Software Engineer*

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SUMMARY

Senior Software Engineer with 10+ years of experience building scalable SaaS platforms, distributed backend systems, and data-intensive infrastructure. Strong background in designing multi-tenant architectures, high-performance APIs, and systems that support large-scale product and enterprise workflows.

Currently building backend systems at Jedify, focusing on scalable SaaS infrastructure and AI-assisted workflow systems. Previously worked at Airtable on core database systems, monetization infrastructure, and enterprise-scale platform architecture. Earlier experience includes engineering and research roles at Google, Microsoft Research, Quora, and Dropbox. Strong systems thinker with expertise in backend architecture, performance optimization, and large-scale distributed system design.

TOP SKILLS

Languages

Python, Java, JavaScript, TypeScript, C++, Go, SQL, Bash

Databases

PostgreSQL, MySQL, Redis, SQLite, MongoDB (familiar), DynamoDB (familiar), indexing strategies, query optimization, sharding, replication

Cloud & Infrastructure

AWS (EC2, S3, RDS, Lambda), Docker, CI/CD pipelines, Terraform (IaC), cloud architecture patterns

AI / Data

LLM API integration, workflow automation systems, data-driven product features

Backend / API Development

REST APIs, Microservices, GraphQL (familiar), gRPC (familiar), event-driven systems, API design, async processing

System Design / Architecture

Large-scale backend systems, low-latency services, high-throughput systems, scalability design, reliability engineering

Security

Authentication, authorization (RBAC), OAuth2, secure API design, encryption (in transit/at rest), rate limiting

Collaboration

Cross-functional engineering, system design reviews, code reviews, product engineering collaboration

EXPERIENCE

Senior Software Engineer, Jedify

09/2023 – Present | Lake Forest, CA

- Architected production backend services that integrate AI capabilities into Jedify's enterprise data and knowledge workflows without developing or training foundation models.

- Built secure ingestion and synchronization pipelines for structured sources such as databases, warehouses, BI tools, and systems of record, plus unstructured knowledge from documents and collaboration tools.
- Integrated managed AI and model-provider APIs behind a provider-agnostic service layer, adding request validation, routing, timeouts, retries, fallbacks, and usage controls for reliable product execution.
- Developed FastAPI services that connect conversational analytics and data-agent experiences to business context, semantic definitions, and governed enterprise data.
- Implemented context-retrieval flows that combine metadata, business rules, semantic entities, source provenance, and user permissions before AI-generated responses reach product surfaces.
- Built backend integration paths for MCP-compatible clients and application SDKs while preserving entity-level authorization and preventing unnecessary exposure of raw schemas.
- Designed asynchronous processing for connector ingestion, document analysis, semantic updates, and long-running research workflows with idempotency, checkpoints, and recoverable retries.
- Optimized PostgreSQL, MongoDB, and Redis access patterns and caching for repeated context lookups, workflow state, and high-frequency API operations.
- Containerized and deployed services on AWS using Docker, Kubernetes/EKS, Helm, S3, RDS, ElastiCache, and managed AI services such as Bedrock.
- Established production observability through structured logging, metrics, health checks, alerting, and failure diagnostics across APIs, connectors, background jobs, and external AI dependencies.
- Partnered with product, data, and customer-facing teams to translate enterprise analytics use cases into reusable integrations for MCP, Slack, Looker, embedded analytics, and custom applications

Tech Stack: Python 3.11+, FastAPI, PostgreSQL, MongoDB, Redis, LiteLLM, REST APIs, AWS (EKS, S3, Bedrock, ElastiCache, RDS), Docker, Kubernetes, Helm

Software Engineer, *Airtable*

10/2019 – 09/2023 | San Francisco

- Worked on core in-memory database layer powering Airtable's real-time collaborative SaaS platform
- Designed backend systems supporting multi-tenant workspace isolation and large-scale data access patterns
- Built and scaled growth infrastructure supporting user acquisition tracking, activation funnels, and experimentation systems
- Led monetization systems including subscription infrastructure, billing workflows, and enterprise pricing models
- Developed backend services supporting enterprise onboarding and workspace provisioning at scale
- Improved query execution performance through indexing strategies, caching layers, and data access optimization
- Contributed to system architecture for scaling enterprise adoption across large organizations
- Ran the first set of Growth experiments and launched education plans, email-based discounts, and extended trials designed for larger teams.

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Tech Stack: Python, Go, TypeScript, PostgreSQL, Redis, distributed systems

Head of Product

06/2017 – 10/2019 | San Francisco Bay Area

Kandra Labs (creators of Zulip)

- Co-built Zulip into a widely used open-source enterprise communication platform
- Designed backend systems including analytics, billing, and enterprise feature infrastructure
- Defined scalable architecture enabling external contributors to extend platform functionality
- Built core product systems supporting real-time messaging and collaboration workflows
- Led product direction including roadmap planning, prioritization, and system design decisions
- Managed external communication including support, sales inquiries, and documentation strategy
- Improved onboarding and adoption workflows for enterprise and open-source users
- Supported early-stage funding efforts including NSF grant acquisition

Tech Stack: Python, Django, PostgreSQL, JavaScript

Member, South Park Commons

06/2016 – 06/2018 | San Francisco Bay Area

- Selected member of technical founder/engineer residency community
- Collaborated with early-stage founders on product and system design thinking
- Participated in technical discussions around scaling startups and product architecture

Software Engineer, Kandra Labs (Zulip)

06/2016 – 06/2017

- First engineering hire contributing to core backend architecture and system design
- Built foundational messaging and collaboration infrastructure
- Implemented early analytics and usage tracking systems
- Contributed to open-source architecture enabling external developer ecosystem
- Worked directly with founders on product iteration and system development

Research Intern, Google

06/2015 – 09/2015

Greater New York City Area

Worked on query clustering and intent modeling research for large-scale search understanding systems.

Research Intern, Microsoft

06/2014 – 09/2014 | Bengaluru Area, India

Researched theoretical approximation bounds for combinatorial optimization problems such as set cover.

Data Science Intern, Quora

06/2012 – 09/2012 | San Francisco Bay Area

Worked on ranking and recommendation experiments for answer quality and topic discovery systems.

Analytics Intern, Dropbox

06/2011 – 09/2011 | San Francisco Bay Area

Built longitudinal analysis framework to analyze user behavior and retention patterns.

EDUCATION

Doctor of Philosophy - PhD, Computer Science , <i>Stanford University</i>	2011 – 2016
Master of Engineering - MEng, Computer Science <i>Massachusetts Institute of Technology</i>	2010 – 2011
Bachelor of Science - BS, Mathematics and Computer Science <i>Massachusetts Institute of Technology</i>	2008 – 2011